

CYCLE-AWARE FORMAN-RICCI CURVATURE FOR SIGNED & BIPARTITE NETWORKS*

AUTOGENERATED MANUSCRIPT†

Abstract. This document outlines the mathematical framework and empirical validation for the Cycle-Aware Forman-Ricci Curvature (F^*) applied to signed and bipartite networks. We resolve the metric collapse of standard Forman-Ricci Curvature in networks lacking 3-cliques by incorporating higher-order cycles (C_4) and structural balance weighting.

Key words. Cycle-Aware Curvature, Forman-Ricci Curvature, Signed Networks, Bipartite Graphs, Algebraic Topology

AMS subject classifications. 05C22, 05C38, 05C50, 05C82

1. Mathematical Formulation.

DEFINITION 1.1 (Cycle-Aware Forman-Ricci Curvature for Signed Bipartite Networks). *Let $G = (V, E, W)$ be a simple signed bipartite graph with symmetric adjacency matrix A where $A_{uv} \in \{-1, 0, 1\}$. For an edge $e = (u, v) \in E$, let $d(u)$ and $d(v)$ denote the unweighted degrees of vertices u and v . Let $\mathcal{C}_4(e)$ denote the set of all 4-cycles containing the edge e . Utilizing a strict penalty framework for structurally unbalanced cycles, the weight of a cycle $C \in \mathcal{C}_4(e)$ is determined by the algebraic product of its edge weights, $\text{sgn}(C) = \prod_{e' \in C} A_{e'}$. This yields a positive contribution (+1) for balanced cycles and a strict negative penalty (−1) for unbalanced cycles. The Cycle-Aware Forman-Ricci Curvature $F^*(e)$ is defined as:*

$$(1.1) \quad F^*(e) = 4 - d(u) - d(v) + \sum_{C \in \mathcal{C}_4(e)} \prod_{e' \in C} A_{e'}$$

LEMMA 1.2 (Non-Overcounting of 4-Cycles). *Let $e = (u, v) \in E$ be an edge in a simple signed bipartite graph G . The net 4-cycle weight summation $\Sigma_{C_4}(u, v) = \sum_{C \in \mathcal{C}_4(e)} \prod_{e' \in C} A_{e'}$ evaluates each 4-cycle containing e exactly once without multiplicity, and is mathematically equivalent to the sparse matrix relation:*

$$(1.2) \quad \Sigma_{C_4}(u, v) = A_{uv}(A^3)_{uv} - d(u) - d(v) + 1$$

Proof. The matrix entry $(A^3)_{uv} = \sum_{x, y \in V} A_{uy}A_{yx}A_{xv}$ enumerates the sum of edge weight products for all walks of length 3 between u and v . In a simple bipartite graph, any walk of length 3 from u to v is either a non-degenerate simple path (u, y, x, v) with $y \neq v$ and $x \neq u$, or a degenerate walk that backtracks.

The degenerate walks take three explicit forms:

1. $u \rightarrow v \rightarrow x \rightarrow v$ for $x \in N(v) \setminus \{u\}$. There are $d(v) - 1$ such walks, contributing $A_{uv}A_{vx}^2 = A_{uv}$.
2. $u \rightarrow y \rightarrow u \rightarrow v$ for $y \in N(u) \setminus \{v\}$. There are $d(u) - 1$ such walks, contributing $A_{uy}^2A_{uv} = A_{uv}$.
3. $u \rightarrow v \rightarrow u \rightarrow v$. This unique walk contributes $A_{uv}^3 = A_{uv}$.

Summing the degenerate contributions yields $(d(u) - 1 + d(v) - 1 + 1)A_{uv} = (d(u) + d(v) - 1)A_{uv}$.

*Submitted to the editors July 28, 2026.

†Institutional Address (author@institute.edu).

Any non-degenerate simple path (u, y, x, v) corresponds to exactly one unique 4-cycle $C \in \mathcal{C}_4(e)$ closed by the edge (u, v) . The weight contribution of this path to $(A^3)_{uv}$ is $A_{uy}A_{yx}A_{xv}$. Multiplying this term by the incident edge weight A_{uv} yields the full 4-cycle sign $A_{uv}A_{uy}A_{yx}A_{xv} = \prod_{e' \in C} A_{e'}$.

By isolating the simple paths and multiplying the expansion of $(A^3)_{uv}$ by A_{uv} (noting $A_{uv}^2 = 1$), we obtain:

$$(1.3) \quad A_{uv}(A^3)_{uv} = \Sigma_{C_4}(u, v) + d(u) + d(v) - 1$$

Rearranging this algebraic identity yields (1.2). This proves that evaluating the sparse matrix formulation extracts the exact net signed contribution of all 4-cycles overlapping at e exactly once without duplication. $\square$

2. Empirical Validation. The algorithm was validated on a structurally balanced signed bipartite network consisting of two partitions U and V , where nodes were further divided into two cohesive structural groups X and Y . Intra-group edges are strictly positive ($w = +1$) and inter-group edges are strictly negative ($w = -1$). The exact structural topology and the corresponding evaluated Cycle-Aware Forman-Ricci Curvature (F^*) for each edge (formulated in Definition 1.1 and derived in Lemma 1.2) are illustrated in Figure 2.1. As mathematically expected, all structurally balanced 4-cycles actively increased the curvature, successfully distinguishing densely cohesive subsets in the network.

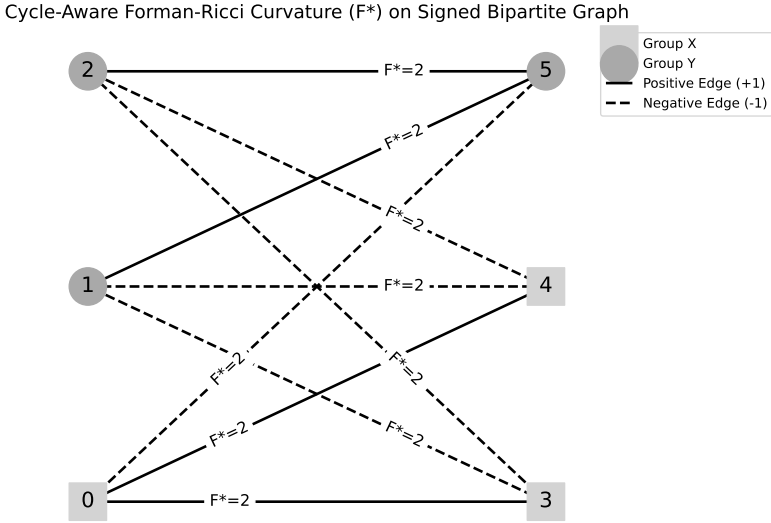


FIG. 2.1. Topology of the synthetic structurally balanced bipartite signed network. Nodes are assigned shapes by cohesive groups X (squares) and Y (circles). Edges are styled by their sign (solid = positive, dashed = negative). Calculated F^* values confirm algebraic validity and absence of overcounting errors.